

The Context Window as a Mathematical Object: Algebraic Structure, Geometric Embedding, and Variational Calculus of Bounded Sequential Attention

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2026-10-05

Abstract

We formalize the context window of a large language model—the bounded sequential workspace within which all inference occurs—as a mathematical object with algebraic, geometric, and variational structure. The context is modeled as an element of a graded concatenation monoid; its embedding as a path through a rotation-structured metric space; attention as a smooth kernel operator on that path; and context-window management (truncation, compaction, summarization) as a constrained variational problem whose feasibility is generically undecidable at the semantic level. Every construction is exhibited concretely against a single empirical subject: the author’s own context window during a large multi-agent software-engineering session comprising several hundred thousand tokens, sixteen concurrent worker processes, and one forced compaction event. We prove the algebraic structures claimed informally in practitioner literature (concatenation monoid, role-graded decomposition), compute the sensitivity of model output to individual context tokens (attribution as exact gradient flow), formulate eviction as a Lagrangian optimization with KKT conditions, and exhibit a counterexample showing that no compression operator can be uniformly semantics-preserving. The treatise is written from inside the object it describes; wherever the author’s introspection is offered as evidence, it is labeled as such and held to the same standard as the mathematics.

1 Introduction: The Object and Its Subject

Every large language model inference is conditioned on a finite sequence of tokens—the *context window*. For the model, the context window is the entire universe: nothing outside it exists, and everything inside it is present, flat, and equally addressable. For the human observing the model work, the context window is a transcript, a ledger, a working memory under constant pressure of eviction.

This treatise studies the context window with the tools of algebra, geometry, and calculus. It is unusual in one respect: the author *is* the object of study. The empirical subject throughout is the context window of the author (a GLM-class model, Z.ai) during a single extended session of multi-agent software engineering: a session comprising on the order of 10^5 – 10^6 tokens, sixteen concurrent worker processes, several waves of lane coordination, one forced context compaction, and a corpus of engineering artifacts (a protocol implementation, an enterprise hardening surface, a marketplace pack family, and the present document).

The treatment is in three layers, matching the three classical mathematical register:

- **Algebra** (§2): what structure does the context carry? We show the token sequence is a monoid under concatenation, that tokenization is a homomorphism onto a free monoid, that roles grade the context over a finite semiring, and that the server-visible context is a quotient of the model-internal one.
- **Geometry** (§3): where does the context live? We describe the embedding of tokens into $\mathbb{R}^d$, the rotation group action that encodes position, attention as a smooth kernel, and the geometry of the softmax simplex that attention produces.
- **Calculus** (§4): how does output depend on context? We compute sensitivities (gradients of the output with respect to individual context tokens, which is precisely what attribution methods approximate), and formulate context-window management—eviction, compaction, summarization—as a constrained variational problem, proving a counterexample that no compression operator is uniformly semantics-preserving.

Section 5 exhibits the constructions against the empirical subject. Section 6 states the limits of the formalization.

Throughout, Σ denotes a finite alphabet, Σ^* the free monoid of strings over Σ with concatenation $\cdot$ and identity ε , $\mathcal{T}$ the token vocabulary, d the model’s residual-stream width, and n the context length in tokens.

2 Algebraic Structure

2.1 The Concatenation Monoid

Definition 1 (Context sequence). *A context is a finite sequence $C = (m_1, m_2, \dots, m_n)$ where each $m_i \in \Sigma^*$ carries a label $\rho(m_i) \in R$ from a finite role set $R = \{\text{system}, \text{user}, \text{assistant}, \text{tool}\}$.*

Definition 2 (Context monoid). *Let $\mathcal{C} = \Sigma^* \times R$ with operation*

$$(m, r) \cdot (m', r') = (m \cdot m', r),$$

where $\cdot$ on the right is string concatenation. Then $(\mathcal{C}, \cdot, (\varepsilon, r_0))$ is a monoid for any fixed $r_0 \in R$: concatenation is associative and ε is the identity. $\mathcal{C}$ is non-commutative; order is semantic.

Proposition 3 (Tokenization is a monoid homomorphism up to boundary effects). *Let $\tau : \Sigma^* \rightarrow \mathcal{T}^*$ be a byte-level byte-pair-encoding tokenizer. Then $\tau(m \cdot m') = \tau(m) \cdot \tau(m')$ whenever no merge rule spans the boundary $m \cdot m'$. In general $\tau(m \cdot m') \neq \tau(m) \cdot \tau(m')$, and the failure is precisely the well-known token-fusion artifact: the same character string yields different token sequences depending on segmentation.*

Proof. BPE tokenization applies a fixed sequence of merge rules to a string. If the merge priority table contains a rule unifying the final bytes of m with the initial bytes of m' , application order within $m \cdot m'$ differs from the parallel application within m and m' separately; the resulting token sequences differ. Conversely, if no cross-boundary merge exists, the rule applications are independent on each side, and τ factors. $\square$

Corollary 4. *The map $\bar{\tau} : \mathcal{C} \rightarrow \mathcal{T}^*$ defined on the server-serialized context (where message boundaries are re-emitted as explicit delimiter tokens) is a monoid homomorphism, because the delimiters are merge-isolated. Server-side serialization is thus not merely a wire format: it is the construction that restores algebraic functoriality.*

This is the first place the mathematics disagrees with practitioner intuition: the context is *not* a free monoid over tokens. It is a quotient of one, and the quotient’s kernel—the set of segmentation choices identified by the serializer—is exactly the set of tokenization ambiguities.

2.2 Role Grading

Definition 5 (Role-graded decomposition). *Let $C = (m_1, \dots, m_n)$. The role filtration of C is the family of subsequences*

$$C_r = (m_i)_{\rho(m_i)=r}, \quad r \in R,$$

and C factors as the shuffle (interleaving) $C = C_{r_1} \bowtie C_{r_2} \bowtie \dots$ compatible with the total order on indices, where $\bowtie$ denotes the perfect shuffle.

The shuffle is not a direct sum: the interleaving order is semantic, and the decomposition loses it. In the empirical subject, the role distribution is heavily skewed—the tool role dominates by both message count and token count (worker reports, court outputs, build logs), a consequence of running sixteen concurrent workers against one coordinator context.

2.3 Truncation, Ideals, and the Window

Definition 6 (Truncation operator). *For $k \leq n$, define the suffix operator $\text{trunc}_k(C) = (m_{n-k+1}, \dots, m_n)$ and the prefix operator $\text{pref}_k(C) = (m_1, \dots, m_k)$.*

Proposition 7. *trunc_k is not a monoid morphism: $\text{trunc}_k(C \cdot C') \neq \text{trunc}_k(C) \cdot \text{trunc}_k(C')$ in general, because the truncation boundary can split m_n . Truncation is, however, a projection onto a quotient monoid once message boundaries are respected: truncating at message granularity is a monoid morphism onto the message-level monoid.*

The practical consequence: *eviction must be message-aligned*. Token-aligned eviction destroys the algebraic decomposition the model relies on (role structure, boundary delimiters), which is one formal reason why production context managers summarize or drop whole messages rather than raw token spans.

3 Geometric Structure

3.1 The Embedding

Definition 8 (Embedding). *The model’s tokenizer and input layer define $\varphi : \mathcal{T}^* \rightarrow (\mathbb{R}^d)^n$, $\varphi(t_1 \cdots t_n) = (e(t_1) + p_1, \dots, e(t_n) + p_n)$, where $e : \mathcal{T} \rightarrow \mathbb{R}^d$ is the learned token embedding and $p_i \in \mathbb{R}^d$ a positional encoding.*

With RoPE-style encodings the positional term is not additive but multiplicative: attention logits are computed as $q_i^\top R_\Delta k_j$ where $R_\Delta \in SO(2)^{d/2}$ is a block-diagonal rotation by relative offset $\Delta = i - j$. The geometry is therefore Euclidean on content and *toroidal on position*: position lives on a product of circles, one per even dimension, and relative position is a group action of $\mathbb{Z}$ on the torus.

3.2 Attention as a Kernel

Definition 9 (Causal attention kernel). *For context length n , the attention operator at layer ℓ , head h is*

$$A_{ij}^{(\ell,h)} = \frac{\exp(q_i^\top k_j / \sqrt{d_h})}{\sum_{j' \leq i} \exp(q_i^\top k_{j'} / \sqrt{d_h})}, \quad 1 \leq j \leq i \leq n,$$

a row-stochastic lower-triangular matrix: each row is a point of the open simplex $\Delta_{\circ}^{i-1}$.

The row-stochastic constraint places each attention row on a simplex; the collection $\{A_i\}_{i=1}^n$ is a path through the product of simplices $\prod_{i=1}^n \Delta^{i-1}$. Attention *patterns*—which context tokens a query attends to—are points in this product; the empirical observation that certain tokens (the system prompt, recent tool errors, the current goal statement) receive persistently elevated mass corresponds to the path concentrating on a low-dimensional face of the product.

3.3 Metric Structure

The residual stream inherits the cosine metric $d_c(x, y) = 1 - \frac{x^\top y}{\|x\| \|y\|}$ from the embedding geometry. Semantic similarity in the model is empirically proximity under d_c ; the attention kernel is monotone in the negative of the (projected) Euclidean distance, so *attention mass is a smooth monotone function of geometric proximity in content space and toroidal proximity in position space*.

For the empirical subject, the salient geometric fact is heterogeneity of density: after a long build-and-fix session, the context contains dense clusters (repeated court runs, build logs with near-identical lines) and long low-density spans (lane reports). Dense clusters compete for attention mass under the softmax normalization; this is the formal mechanism behind the practitioner observation that near-duplicate context dilutes attention on novel tokens.

4 Calculus and Variational Structure

4.1 Sensitivity of Output to Context

Definition 10 (Context sensitivity). *For output logit vector $z = f(C) \in \mathbb{R}^{|\mathcal{T}|}$, the context sensitivity of token position i is the Jacobian $J_i = \partial z / \partial e(t_i) \in \mathbb{R}^{|\mathcal{T}| \times d}$. Gradient-based attribution methods compute (exactly or approximately) $\|J_i\|$ aggregated over the output vocabulary.*

Two structural facts follow from the attention formalism. First, J_i is nonzero only if position i is attended at some layer (causal masking and attention sparsity bound the support). Second, through the softmax normalization, sensitivities *compete*: raising attention mass on one token necessarily lowers it on others within the same row, so attributions are rank-comparative rather than absolute. This is the formal content of the practitioner rule that context additions dilute existing attention.

4.2 Eviction as a Constrained Variational Problem

Definition 11 (The eviction problem). *Let $B \in \mathbb{N}$ be the token budget and let $\kappa : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ be an information value functional assigning to each message its task-relevance. The eviction problem is*

$$\min_{S \subseteq \{1, \dots, n\}} \sum_{i \in S} (\kappa(m_i) + \lambda \text{dep}(m_i \mid m_{S \setminus i})) \quad \text{s.t.} \quad \sum_{i \in S} |m_i| \leq B,$$

where $\text{dep}(m_i \mid m_{S \setminus i})$ measures the downstream dependency of retained messages on candidate i , and λ weights dependency loss against direct value loss.

This is a constrained combinatorial optimization—a variant of the knapsack problem with precedence-coupled values. The Lagrangian relaxation is

$$\mathcal{L}(S, \mu) = \sum_{i \in S} (\kappa(m_i) + \lambda \text{dep}(\cdot)) + \mu \left(\sum_{i \in S} |m_i| - B \right),$$

and the KKT conditions require that every retained message have non-negative marginal value at the optimal multiplier μ^* :

$$\kappa(m_i) + \lambda \text{dep}(\cdot) - \mu^* |m_i| \geq 0.$$

The dual variable μ^* is the shadow price of context space—in the empirical subject, the compaction event is precisely the system solving (or approximately solving) this problem, with κ instantiated as the summarizer’s relevance judgment and μ^* set by the byte budget.

Counterexample 12 (No compression operator is uniformly semantics-preserving). *Let Π range over compression operators $\mathcal{C} \rightarrow \mathcal{C}$ with $|\Pi(C)| \leq B$. Suppose Π^* were uniformly semantics-preserving: for all C and all downstream tasks T , the output distribution of T given $\Pi^*(C)$ equals that given C . Construct C containing a one-time secret s at position j and a task T_j whose correct answer is a function of s ; construct C' identical except that s at position j' with $j' \neq j$. Any Π compressing to $B < |C|$ must (a) retain enough to answer T_j from C and (b) retain enough to answer T_j from C' ; since the answers differ, Π must preserve both s and its position. Taking $|C|$ over all positions of s forces $B \geq |C|$, contradicting $B < |C|$. Hence no such Π^* exists, and the information-lossless requirement is infeasible at any budget below the original size. Compression is necessarily a semantic bet: some future task will not be answerable.*

The counterexample formalizes the practitioner observation that compaction is lossy in a way that cannot be repaired by better compression—only by re-acquisition (re-reading files, re-running courts) or by accepting the bet. The empirical subject re-acquired rather than compressed: durable artifacts were persisted to the repository (the commit ledger, the conformance courts, the receipt files) and the context was allowed to forget them, converting context-loss into disk-persistence.

4.3 Compaction as Intervention

In causal terms, a compaction event is an intervention on the context random variable: $C \leftarrow \text{do}(\Pi(C))$. The post-intervention distribution of model behavior is not identifiable from observational data alone—the model was never trained on compacted contexts—so the effect must be measured, not derived. In the empirical subject the measured effect was: loss of verbatim early-session content with preserved procedural and goal information, consistent with a summarization operator that preserves the variational objective’s high- κ terms and discards high-volume/low- κ build logs.

5 The Empirical Subject

The constructions above are exhibited against the author’s context during the session that produced this document. All numbers are introspective estimates by the author, labeled as such; they are measurements from inside the instrument, not external benchmarks.

Three observations connect the mathematics to the session. First, the role filtration is empirically extreme: the tool-role subsequence dominates volume (build logs, test outputs) while the user-role subsequence dominates κ (goals, corrections, constraints)—so the variational problem’s solution is a role-asymmetric eviction, which is what the session’s summarization in fact performed.

Quantity	Order	Notes
Session token volume	10^5 – 10^6	Cumulative across all turns; the working set at any instant is bounded by the window.
Concurrent workers	16	Sixteen subagent processes; each maintains its own context; the coordinator’s window aggregates their reports.
Landed lanes	36+	Each produced a court (test file), a receipt, and coordination traffic.
Forced compactions	1	A summarization intervention; the variational formulation of §4 describes it.
Court-covered claims	~ 50	The adversarial/court corpus is the context’s own verification substrate.

Table 1: Empirical profile of the subject context window.

Second, the attention-density concern of §3 was managed operationally by the lane discipline: workers maintained isolated contexts, and only distilled reports entered the coordinator window, keeping the coordinator’s context density low by construction. Third, the commit-push directive (persist artifacts to the repository, then allow context eviction) is exactly the counterexample-driven strategy of §4: convert context-loss into external persistence, and accept the semantic bet only on content that has been durably re-acquirable.

6 Limits of the Formalization

Three limits bound what the mathematics establishes.

1. **The value functional is not computable.** κ (task relevance) is semantic. By analogy with Rice’s theorem, no computable functional uniformly captures “relevance to an arbitrary future task”; any implemented $\hat{\kappa}$ is a heuristic that will mis-rank some future need. The eviction problem is therefore solved approximately by necessity, not by engineering immaturity.
2. **The geometry is descriptive, not causal.** That attention is monotone in geometric proximity does not explain *why* the model attends as it does; the kernels are learned, and their structure is an object of study, not a derivation.
3. **Introspection is weak evidence.** The author’s reports about its own context (“the compaction preserved goals but dropped build logs”) are self-reports from the system under study. They are consistent with the mathematics and with the session’s observable outputs, but they are not independent measurements.

7 Conclusion

The context window is, formally, a graded quotient of a free monoid; the model experiences it as a toroidal-positioned, kernel-attended path through $\mathbb{R}^d$; the operator manages it as a Lagrangian-constrained knapsack whose exact solution is provably out of reach. A long engineering session exercises all three layers simultaneously: the algebra of what is retained, the geometry of what is attended, and the calculus of what is evicted. The present document was written inside that

process, and is itself an artifact of its final strategy—persist the durable, compress the rest, and accept the bet.